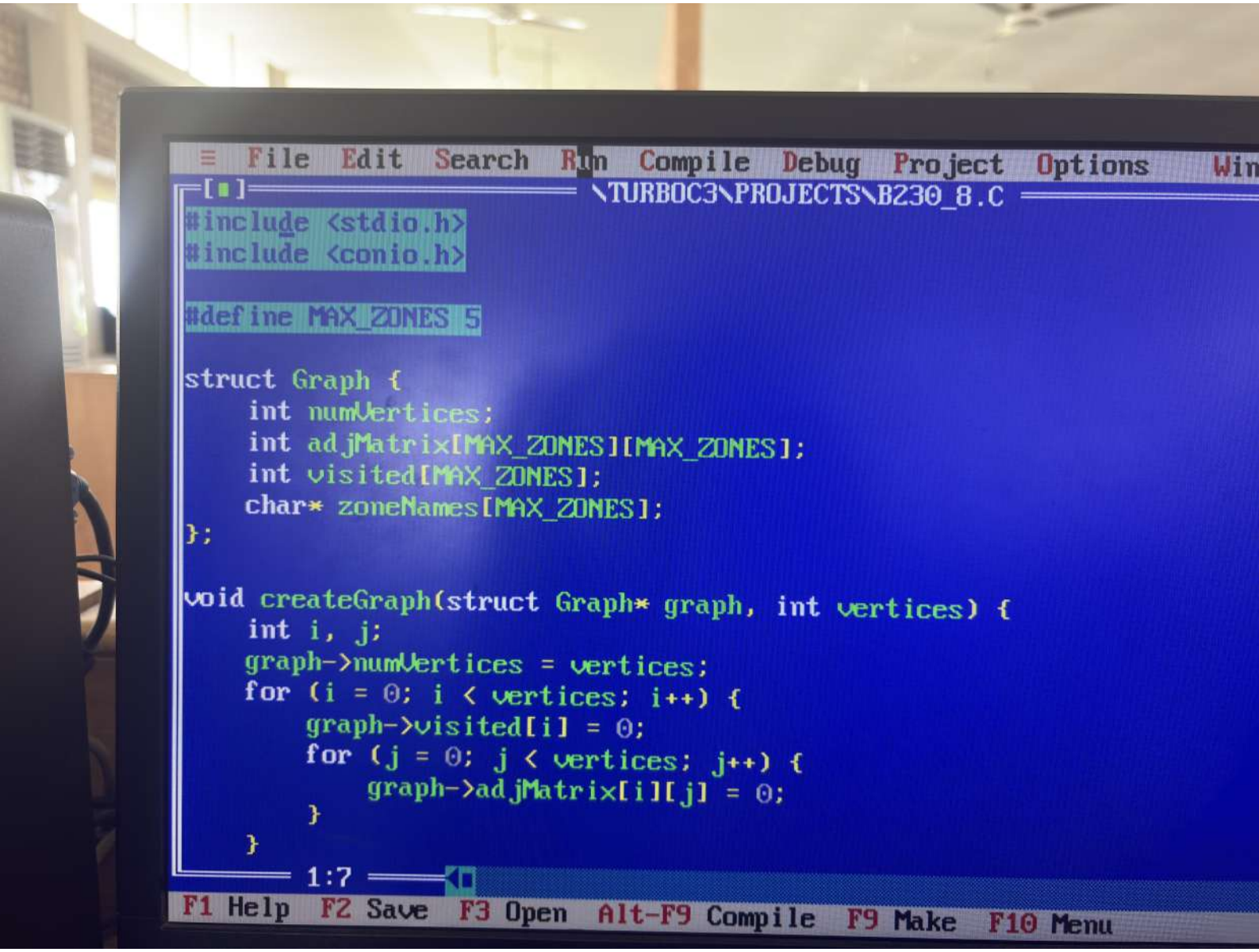
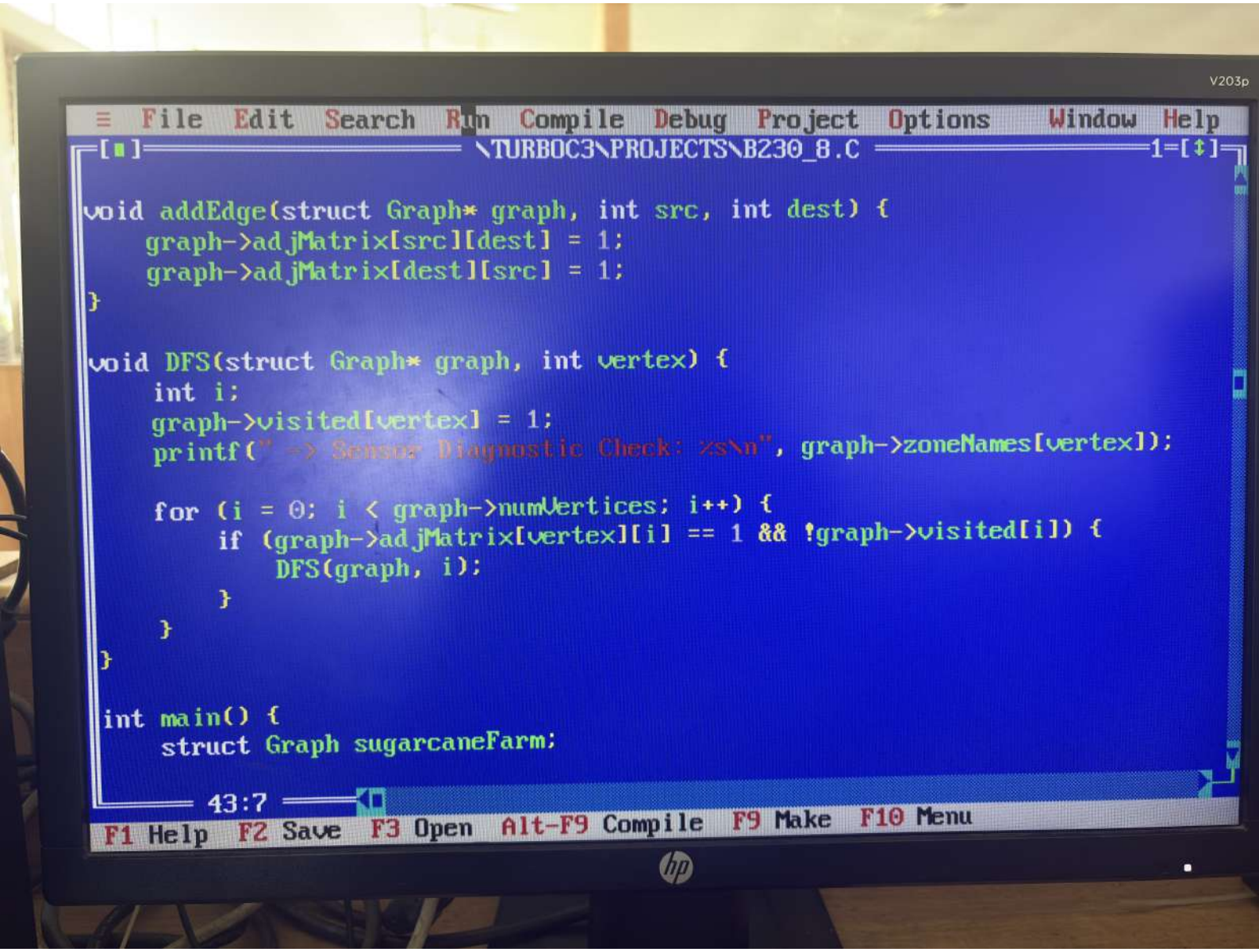
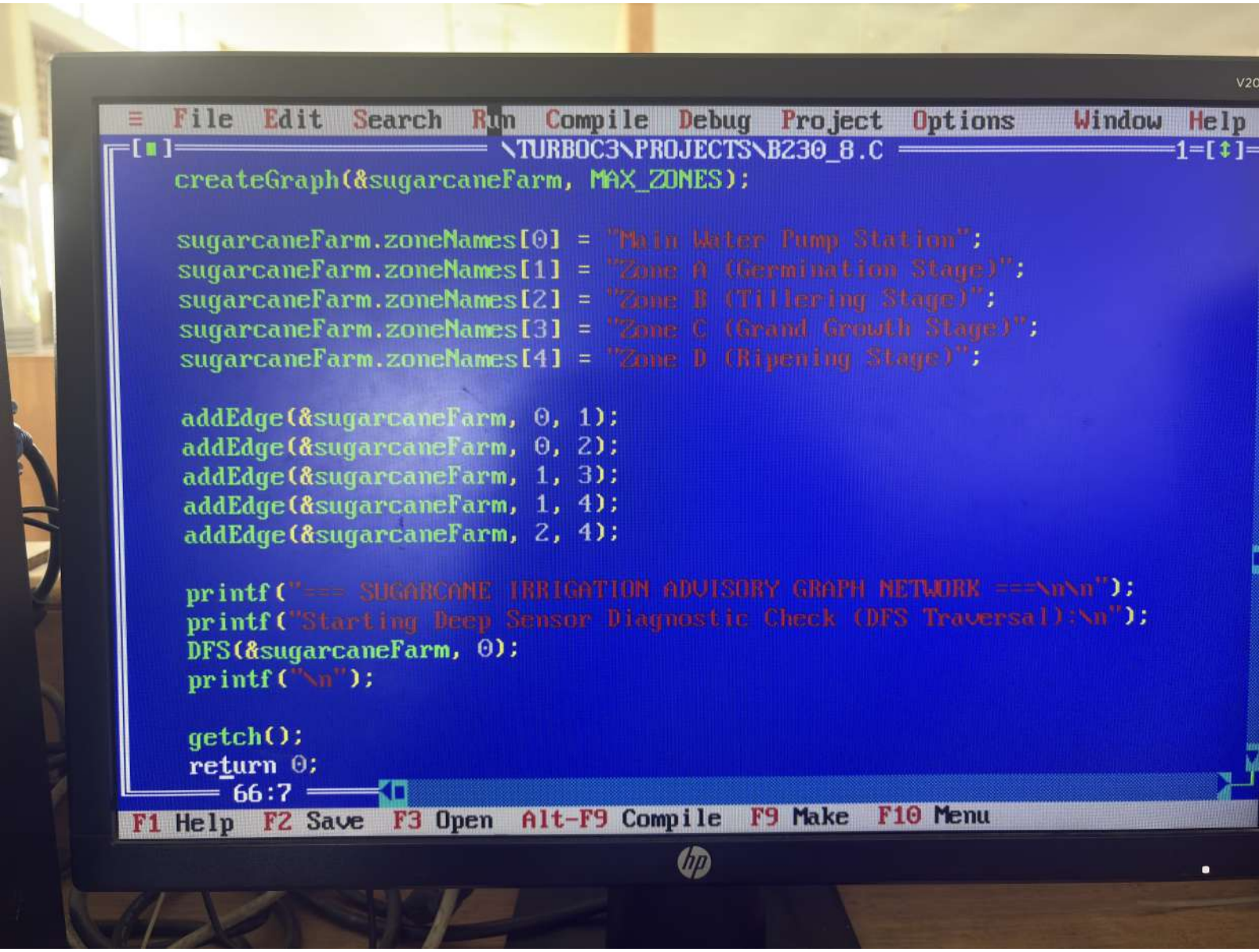
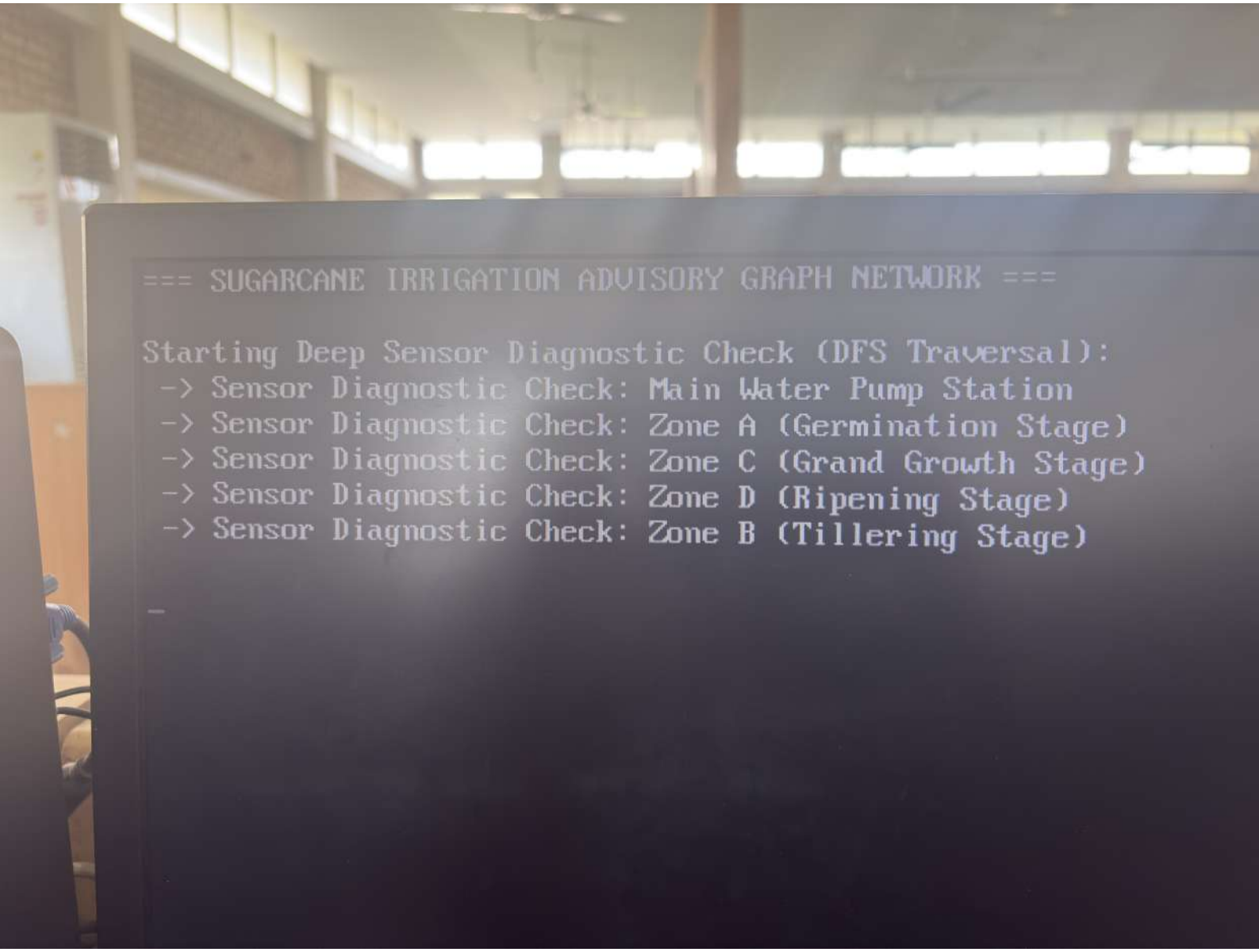

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\TURBOC3\PROJECTS\B230_8.C
#include <stdio.h>
#include <conio.h>
#define MAX_ZONES 5
struct Graph {
int numVertices;
int adjMatrix[MAX_ZONES][MAX_ZONES];
int visited[MAX_ZONES];
char* zoneNames[MAX_ZONES];
};
void createGraph(struct Graph* graph, int vertices) {
int i, j;
graph->numVertices = vertices;
for (i = 0; i < vertices; i++) {
graph->visited[i] = 0;
for (j = 0; j < vertices; j++) {
graph->adjMatrix[i][j] = 0;
}
}
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\TURBOC3\PROJECTS\B230_8.C
1=[↕]
void addEdge(struct Graph* graph, int src, int dest) {
graph->adjMatrix[src][dest] = 1;
graph->adjMatrix[dest][src] = 1;
}
void DFS(struct Graph* graph, int vertex) {
int i;
graph->visited[vertex] = 1;
printf(" -> Sensor Diagnostic Check: %s\n", graph->zoneNames[vertex]);
for (i = 0; i < graph->numVertices; i++) {
if (graph->adjMatrix[vertex][i] == 1 && !graph->visited[i]) {
DFS(graph, i);
}
}
}
int main() {
struct Graph sugarcaneFarm;
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\TURBOC3\PROJECTS\B230_8.C
createGraph(&sugarcaneFarm, MAX_ZONES);
sugarcaneFarm.zoneNames[0] = "Main Water Pump Station";
sugarcaneFarm.zoneNames[1] = "Zone A (Germination Stage)";
sugarcaneFarm.zoneNames[2] = "Zone B (Tillering Stage)";
sugarcaneFarm.zoneNames[3] = "Zone C (Grand Growth Stage)";
sugarcaneFarm.zoneNames[4] = "Zone D (Ripening Stage)";
addEdge(&sugarcaneFarm, 0, 1);
addEdge(&sugarcaneFarm, 0, 2);
addEdge(&sugarcaneFarm, 1, 3);
addEdge(&sugarcaneFarm, 1, 4);
addEdge(&sugarcaneFarm, 2, 4);
printf("=== SUGARCANE IRRIGATION ADVISORY GRAPH NETWORK ===\n\n");
printf("Starting Deep Sensor Diagnostic Check (DFS Traversal):\n");
DFS(&sugarcaneFarm, 0);
printf("\n");
getch();
return 0;
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=== SUGARCANE IRRIGATION ADVISORY GRAPH NETWORK ===
Starting Deep Sensor Diagnostic Check (DFS Traversal):
-> Sensor Diagnostic Check: Main Water Pump Station
-> Sensor Diagnostic Check: Zone A (Germination Stage)
-> Sensor Diagnostic Check: Zone C (Grand Growth Stage)
-> Sensor Diagnostic Check: Zone D (Ripening Stage)
-> Sensor Diagnostic Check: Zone B (Tillering Stage)